



THOMPSON RIVERS UNIVERSITY

SENG 4660 – Agile Game Development

Game Design Document & Playtesting Report

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Part 1: Game Design Document

1. Game Goals

Player Goals:

- Collect as many points as possible by passing through pipes
- Collect as many power-ups as possible
- Defeat as many enemies as possible
- Achieve the highest combined score (pipes passed + enemies defeated)

Designer Goals:

- Create an infinite gameplay experience
- Ensure progressively increasing difficulty through:
 - Increased speed
 - More frequent enemy spawning
 - More challenging pipe placement
- Introduce a variety of power-ups to enhance gameplay and strategy

2. MDA Framework

Mechanics (Core Systems and Rules)

Flight Control

- Hold the Spacebar to glide upward
- Release to descend due to gravity

Combat

- Press Return or Left Mouse Button to attack
- Attacks damage enemy birds

Health System

- Player has a health bar
- Colliding with enemies reduces health
- Colliding with environmental obstacles results in instant defeat

Enemies

- Stationary birds that act as obstacles
- Chasing birds that actively pursue the player

Collectibles

- Hearts: Restore health
- Weapon Upgrade: Temporarily increases attack power
- Repel Power-up: Pushes enemies away for a short duration

Scoring

- Distance traveled
- Enemies defeated
- Collectibles gathered

Dynamics (Gameplay Behavior)

- Players must balance movement and combat simultaneously
- Avoiding pipes while attacking enemies creates constant tension
- Difficulty increases through tighter spacing and increased enemy frequency
- Power-ups encourage risk-taking for higher rewards

Aesthetics (Player Experience)

- Fast-paced aerial survival gameplay
- Sense of skill mastery and progression
- Competitive motivation through score improvement
- Simple yet engaging arcade-style experience

3. Gameplay Overview and Rules

Core Gameplay Loop:

1. The player begins flying forward automatically
2. The player controls altitude using glide mechanics
3. The player avoids pipes and environmental obstacles
4. Enemy birds appear and attempt to block or chase the player
5. The player attacks enemies while maintaining flight
6. The player collects power-ups and health items
7. The score increases based on survival and actions
8. The loop continues until the player loses all health

4. Winning and Losing Criteria

Winning Condition:

- There is no fixed win state
- The objective is to achieve the highest possible score

Losing Condition:

- The game ends when player health reaches zero due to:
 - Enemy collisions
 - Environmental hazards (pipes, ground, etc.)

5. Expected Game Duration

- The game is infinite and continues as long as the player survives
- A session may last anywhere from less than a second to several minutes depending on player skill

6. Technical Considerations

- **Game Engine:** Unity
- **Target Platform:** Mobile devices (Android and iOS)
- **Additional Considerations:**
 - Should run on mobile emulators
 - Optimized for performance on lower-end devices

Part 2: Playtesting Observations

Confusing Rules Observed:

- Gameplay was generally straightforward
- Some confusion between stationary enemies and moving enemies
- Minor confusion regarding the purpose of power-ups

Player Behaviors During Gameplay:

- Players appeared engaged and enjoyed the experience
- Minimal confusion allowed players to focus on gameplay rather than learning mechanics

Unexpected Gameplay Issues:

- No major unexpected issues were observed

Difficulties Experienced:

- No significant difficulties reported during gameplay

Part 3: Player Feedback

What Players Liked:

- Simplicity of the game

- Easy-to-understand controls
- Clear and familiar gameplay style

What Players Found Confusing or Challenging:

- Power-up meanings were not clearly defined

Suggestions for Improvement:

- Include an information screen explaining all power-ups
- Add a clear warning that touching pipes or the ground results in death

Part 4: Change Log

Change 1: Add Detailed Information Menu

- **What:** Introduce a menu explaining power-ups and game rules
- **Why:** To reduce confusion and improve player understanding
- **Source:** Player feedback indicating unclear power-up functionality

Change 2: Add On-Screen Warning or Tutorial Prompt

- **What:** Provide a notice informing players that collisions with pipes or the ground result in death
- **Why:** To ensure players understand critical game rules early
- **Source:** Player suggestion during playtesting

Part 5: Reflection and Conclusion

Through observing players during the playtesting session, we learned that simplicity in design allows players to quickly engage with the game. Most players were able to understand the core mechanics without guidance, which confirmed that our basic gameplay loop is intuitive and effective. However, we also observed that even small gaps in clarity, such as unclear power-up functions, can impact the overall player experience.

Player feedback played a key role in shaping our design decisions. While the core gameplay was well-received, feedback highlighted the importance of better communication with the player. As a result, we decided to implement an information menu and clearer in-game instructions to ensure that all players, including those unfamiliar with this genre, can fully understand and enjoy the game.

For the next iteration, we plan to improve the onboarding experience by adding a tutorial or instructional overlay, refining visual indicators for power-ups, and further balancing difficulty progression. These improvements aim to enhance clarity, accessibility, and overall player satisfaction while maintaining the fast-paced and engaging nature of the game.